Git & GitHub Basics (Beginner Friendly)

Goal: Learn how to upload your project to GitHub using Git Bash.

1. Check Git Installation

- Open Git Bash and type :

```
git --version
```

If you see version number, Git is installed.

2. Configure Git (Once)

- Tell Git your name and email.

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```

- Check configuration : `git config --list`

Do this only once in your system.

3. Go To Your Project Folder

- cd Desktop
- cd MyProject
- Check where you are : `pwd`
- See files : `ls`

4. Initialize Git Repository

```
git init
```

- Meaning : Start tracking this project with Git.

Output : Initialized empty Git repository

5. Check Status

```
git status
```

- Shows :
  - New files
  - Modified files
  - Deleted files

Example :
Untracked files :
index.html
style.css

6. Track Files

- Track one file : `git add index.html`
- Track multiple files : `git add index.html style.css`
- Track everything : `git add .`

Remember : `git add .` means Track all new and modified files.

7. Check Status Again

```
git status
```

Now you'll see :

Changes to be committed

⇒ This means the files are ready to be saved.

8. Commit Your Changes

```
git commit -m "First Commit"
```

- Meaning : Save a snapshot of your project.
- Example Output : [main 9ab23] First Commit

9. Create Repository on GitHub

- Go to GitHub → New Repository
- Enter name (ex : StudentProject)
- Do NOT add README, .gitignore, License
- Click Create Repository.
- Copy the URL shown on the screen.

10. Connect Local Project To GitHub

```
git remote add origin https://github.com/username/StudentProject.git
```

- Check : `git remote -v`

Output :
origin https://github.com/username/StudentProject.git
origin https://github.com/username/StudentProject.git (fetch)

11. Rename Branch

```
git branch -M main
```

- Now your branch is called : main

12. Push Code To GitHub

```
git push -u origin main
```

- Meaning : Upload your project to GitHub.
- Refresh GitHub. Your files will appear. ☺

Daily Workflow (After First Push)

1. Check status → `git status`
2. Add changes → `git add .`
3. Commit changes → `git commit -m "Updated project"`
4. Push to GitHub → `git push`

Complete Workflow (First Time)

`git init` → `git add .` → `git commit -m "First Commit"`
↓ ↓
`git remote add origin URL` → `git branch -M main`
↓
`git push -u origin main`

Important Git Commands

Command	Purpose
git init	Start a new Git repository
git status	Check the status of files
git add .	Track all files
git commit -m "msg"	Commit (save) changes
git remote add origin URL	Connect to remote repository
git remote -v	View remote repository
git branch -M main	Rename branch to main
git push -u origin main	First upload to GitHub
git push	Upload latest changes
git pull	Download latest changes
git clone URL	Copy repository from GitHub
git log	View commit history

Easy Way To Remember

Create Project → `git init`
↓ ↓
`git status` → `git add .`
↓
`git commit -m "First Commit"`

Create GitHub Repository
↓
`git remote add origin URL`
↓
`git branch -M main`
↓
`git push -u origin main`

Memory Trick

A → C → P

- A = git add . Add changes
- C = git commit -m "message" Commit (save a snapshot)
- P = git push Push to GitHub

Every time you update your project, just remember :

```
git add .
git commit -m "Describe your changes"
git push
```

☺